DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #06B: PILLS & ALCHEMY SYSTEM (PART 2 — CULTIVATION-ACCELERATION, DANTIAN/MARROW/BLOODLINE TEMPERING, INSIGHT, SPIRIT-FISH ALCHEMY & LONGEVITY ELIXIRS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Master Roadmap (`06A–06F`)**:
  - `06A`: Group 1 — Basic Welfare, Vital Recovery, Injury-Mending, Detox & Mind-Calming Pills
  - **`06B` (This Document)**: Group 2 — **Cultivation-Acceleration, Dantian/Marrow/Meridian/Bloodline-Tempering, Scripture-Insight, Spirit-Fish Alchemy & Longevity Elixirs (`ITEM190_PILL_006..007`, `REC140_ALC_LONGEVITY_6 / PARTIAL_DRAGON_PILL_7 / TR_ASTRAL_NIRVANA_ELIXIR`, `EFX_DANTIAN_EXPANSION / BONE_MARROW_TEMPER / MERIDIAN_STABILIZE / BLOODLINE_AWAKEN / LIFESPAN_EXTEND`, & `fishing.js`)**
  - `06C`: Group 3 — Realm Breakthrough Pills (`P:BT_*`), Bottleneck Treasures, Resistance & Wrong-Path Toxicity
  - `06D`: Group 4 — Pre-Fight Combat Buff Pills (`P:CB_*`), Tactical/Stealth, Beast-Bond & Forbidden Blood-Sacrifice Pills
  - `06E`: Group 5 — Multi-Phase Specialty Elixirs (`PILL_V234_*`), Elemental Fusion Pills (`PILL_V235_*`) & Experimental Formulas
  - `06F`: Part 6 — Grand Finale: Holistic Overview of the Entire Pill, Toxicity & Alchemy Ecosystem

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`). Never store ad-hoc custom fields inside `npc.h.*`, and remember `SIM.serialize` strips leading-underscore fields (`npc._*`)!

2. **Xianxia Diminishing-Returns Law**: Repeatedly consuming the same Longevity or Marrow-Tempering elixir must exhibit natural diminishing returns (ยาอายุวัฒนะเม็ดแรกเห็นผลดั่งปาฏิหาริย์ เม็ดถัดไปสรรพคุณย่อมลดทอนลงตามกฎธรรมชาติ).

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1. [PILL-GROW-01] Critical Disconnect: 20+ Tempering, Longevity & Bloodline EFX_* Codes Do 0% Effect When Consumed by NPCs via ECO.usePill / ECO.applyPill!

Compare ECO.pillInfo (js/economy.js:325–336, used by **every** autonomous NPC pill pathway: CPE.pickPurchase, CPE.buy, CPE.autoUse, A.personalPillUse, ECO.usePill, ECO.applyPill) against ECO.executeItemEffectById (economy.js:835–1025, called **only** by ECO.useConsumableItem!):

- ECO.pillInfo (lines 325–336) translates **only 13** effectIds (EFX_CULTIVATION_ACCELERATOR, EFX_MASS_VITAL_HEAL, EFX_CALM_MIND, EFX_LEARNING_INSIGHT, EFX_MERIDIAN_TOXIN_PURGE, EFX_PURIFY_HOSTILE_STATUS, EFX_CONTROL_RESILIENCE, EFX_SCENT_TRACE_BREAK, EFX_BEAST_BOND_CHANNEL, EFX_BLOOD_SACRIFICE_OVERDRIVE, and 3 EFX_V234_*).
- **ALL OTHER EFX_* PILL EFFECTS ARE 100% UNMAPPED IN ECO.pillInfo AND NEVER EXECUTED IN ECO.applyPill**, including:
  - EFX_DANTIAN_EXPANSION (ขยายตันเถียนเพิ่มความจุปราณ)
  - EFX_BONE_MARROW_TEMPER (ชุบไขกระดูกขัดเกลากายา)
  - EFX_MERIDIAN_STABILIZE (ประคองและเปิดเส้นลมปราณพิสดาร)
  - EFX_QI_PURITY_REFINEMENT / EFX_CULTIVATION_BURST
  - EFX_BLOODLINE_AWAKEN (ปลุกสายเลือดบรรพกาล)
  - EFX_LIFESPAN_EXTEND / EFX_LONGEVITY_NOURISH (ต่ออายุขัย)
  - EFX_SOUL_NOURISH / EFX_MIND_ENLIGHTEN (บำรุงดวงจิตเพิ่มปัญญา)
- **Result**: Whenever an NPC buys one of these catalog pills with CP and consumes it via CPE.autoUse / A.personalPillUse / ECO.usePill, **ECO.applyPill executes ZERO of its EFX_* effects**!
- **Fix**: Inside ECO.applyPill(W, npc, key) (economy.js:338), if info.item && Array.isArray(info.item.effects), invoke ECO.executeItemEffectById for any effectId not already handled natively by info.effect—and mirror their desirability tags into ECO.pillInfo so CPE.pickPurchase knows when an aging Elder needs EFX_LIFESPAN_EXTEND or a Body cultivator wants EFX_BONE_MARROW_TEMPER!

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2. [PILL-GROW-02] Broken EFX_DANTIAN_EXPANSION, Uncapped EFX_MERIDIAN_STABILIZE & Split-Brain Longevity Variables!

Even when ECO.executeItemEffectById (economy.js:870–915) *is* called directly, three severe implementation bugs corrupt character state:

2.1 npc._dantianBonusPct Is Both Unread by CULT.qiMax AND Deleted on Save/Load (economy.js:886)!

- **Exact Code (economy.js:885–887)**:

```
JavaScript
case 'EFX_DANTIAN_EXPANSION':
  npc._dantianBonusPct = Math.min(0.40, (npc._dantianBonusPct || 0) +
(params.expandPct || 0.06));
```

- **Double Failure**:
  1. CULT.qiMax(npc) in js/cultivation.js **never reads npc._dantianBonusPct** (+0 actual Max Qi!).
  2. Because _dantianBonusPct begins with a leading underscore _, **SIM.serialize deletes it on every Save/Load**!
- **Fix**: Persist in npc.ext.dantianBonusPct (no leading underscore!) and multiply CULT.qiMax(npc) by (1 + clamp(npc.ext?.dantianBonusPct || 0, 0, 0.40))!

2.2 Two Competing Lifespan Variables (c.lifeBonus vs npc.lifespanBonusYears) & Infinite Immortality Stacking (economy.js:349–350 vs 906–910)!

- ECO.applyPill (line 350) adds e.lifespanBonus to **npc.cult.lifeBonus** and calls CULT.refreshLifespan(npc) (which skips Mortals !npc.cult, so a Mortal eating an Immortal Peach gets +0 years!).
- Meanwhile, EFX_LIFESPAN_EXTEND (lines 906–910) writes to **npc.lifespanBonusYears** and mutates **npc.h.life.lifespan** directly without updating c.lifeBonus—so the next time CULT.refreshLifespan(npc) runs (on breakthrough or mantra change), **the EFX_LIFESPAN_EXTEND lifespan gain is overwritten and lost**!
- **And Zero Diminishing Returns**: Eating 15 REC140_ALC_LONGEVITY_6 pills grants 15 * 90 = +1,350 full years with zero resistance!

- **Fix**: Unify all pill lifespan gains into `npc.ext.pillLifespanYears` (affecting both Mortals and Cultivators inside `CULT.refreshLifespan`), with authentic Xianxia diminishing returns per tier (`1st pill: 100%`, `2nd: 55%`, `3rd: 25%`, `4th+: 10%`)—and teach `CPE.pickPurchase` so **aging Elders within `25` years of `lifeMax` prioritize buying Longevity Elixirs above all other items**!

2.3 `EFX_MERIDIAN_STABILIZE` Opens All 8 Extraordinary Meridians at Stage 1 Without Gates (`economy.js:874`)

- `CULT.MERIDIANS.openNextMeridian(npc)` is called unconditionally on every `EFX_MERIDIAN_STABILIZE` pill, letting a Stage-1 novice chug 8 cheap stabilization pills to unlock all 8 Extraordinary Meridians (`acupointImmunity: true`) instantly! Gate pill-assisted meridian opening on `npc.cult.stage >= Math.ceil((openedCount + 1) * 0.75)`.

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3. `[PILL-GROW-03] "starter": true` Bug on Undeciphered Ancient Dragon Pill & Celestial Heritage Nirvana Elixir (`data/data_items.js` & `economy.js:264`)!

Inspect `data/data_items.js` on the two highest-tier Tier-7 alchemy recipes in the game:

1. **`REC140_PARTIAL_DRAGON_PILL_7` (เศษคัมภีร์โอสถหัวใจมังกรอมตะ, Tier 7)**:
   - `descTh`: "ตำรับโบราณที่ขาดหายไปสองส่วน **ยังไม่อาจใช้หลอมจริงได้จนกว่าจะถอดรหัสสำเร็จ**"
   - **Live Bug**: Marked `"starter": true` with `discoverable: undefined`! Every sect automatically knows and brews this "undeciphered incomplete scroll" the instant `recipeTier` hits 7, with zero deciphering required!
2. **`REC140_TR_ASTRAL_NIRVANA_ELIXIR` (มหาตำรับมรดก: โอสถน้ำค้างดารานิรันดร์, Tier 7, `+300` lifespan, `toxicity: 0`)**:
   - `descTh`: "สร้างได้เฉพาะ**ผู้ครอบครองตำรับ**และวัตถุดิบจากมรดกเบื้องบน"
   - **Live Bug**: Also marked `"starter": true` (`discoverable: undefined`)!
3. **Free `+1` Bonus Pill per Batch From `r.starter ? 1 : 0` (`js/economy.js:264` & `309`)!**:

```javascript
const n = (r.yield || 1) + (W.rng.chance(...) ? 1 : 0) + (r.starter ? 1 : 0);
```

Because `"starter": true` was left on Tier 5–7 `REC140_ALC_*` recipes, **1 single `MAT_TR_ASTRAL_DEW` Heritage Relic yields `2–3` Nirvana Elixirs (`+600 to +900 years of lifespan`!)**!

- **Exact Fix**: Set `starter: false, discoverable: true` on `REC140_PARTIAL_DRAGON_PILL_7` (requiring `content.js` scroll restoration!) and `REC140_TR_ASTRAL_NIRVANA_ELIXIR` (unlocked via Heritage Sanctum/Clue!), and restrict the `+ (r.starter ? 1 : 0)` yield bonus in `economy.js:264, 309` strictly to `r.starter && r.tier <= 2`!

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4. `[PILL-GROW-04]` Dead `dailyProgressCap`, Buff Overwriting, Dropped `comprehensionBonus`, & Spirit-Fish Alchemy Deadlock!

1. **Dead `dailyProgressCap: 10` (`ITEM190_PILL_007` โอสถรวมปราณเร่งตบะสามวัฏจักร)**:
   - `dailyProgressCap` in `data_items.js` has **zero references** in any `.js` engine file! Track `npc.ext.dailyPillCultDay` / `dailyPillCultCount` so spamming 10 cultivation pills in 1 hour hits meridian saturation (`diminishing gain + extra toxicity`).
2. **Weak Pill Overwrites Stronger Active `pillBoost` & Drops `masteryGainMultiplier` (`economy.js:328, 342, 351`)**:
   - Line 342 (`c.pillBoost = e.rateBoost`) overwrites a `+0.32` boost with a weaker `+0.15` boost—use `c.pillBoost = Math.max(c.pillBoost || 0, e.rateBoost)` when `c.pillT > 0`!
   - Line 351 (`if (e.insight) npc.insightT = 360;`) throws away `ITEM190_PILL_006`'s `masteryGainMultiplier: 1.45` and `comprehensionBonus: 12`! Store `npc.insightMult = Math.max(1.25, e.insight)` and apply it to `A.study.tick` AND `SK.gainMastery`!
3. **Zero-Qi `SHADOW_SAMURAI` Drinking Pure Qi-Condensation Elixirs**:
   - Gate pure Qi-condensation pills (โอสถน้ำค้างรวมปราณ, โอสถรวมปราณ) for `SHADOW_SAMURAI` (`0.35×` efficiency unless `BODY` / marrow-tempering pill), steering Bushido disciples toward `EFX_BONE_MARROW_TEMPER` and sword-intent tea/training!
4. **Spirit-Fish Alchemy Reactive Visibility Deadlock (`economy.js:88` & `fishing.js`)**:
   - Because `ECO.recipesFor` hides `r.fishAlchemyRecipe` whenever `stock[fishId] === 0`, `ECO.pickOrder` (`line 460`) **never adds Spirit Fish to `W.sect.want`**, so anglers cook rare Spirit Fish into ordinary meals before Alchemists can reserve them! Reserve `1–2` rare Alchemy-grade Spirit

Fish in `ECO.reserveFloor` when `alchemy` hall is active, and verify all `fishAlchemyRecipe` inputs reference catchable T1–T5 fish!

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5. Implementation & Verification Checklist for GPT / Gemini (`06B`)

- [ ] **Bridge All `EFX_*` Pill Effects Into `ECO.pillInfo` & `ECO.applyPill` (`economy.js:325–355`)**: Ensure NPCs consuming catalog pills via `CPE`/`ECO.usePill` receive `EFX_DANTIAN_EXPANSION`, `EFX_BONE_MARROW_TEMPER`, `EFX_BLOODLINE_AWAKEN`, `EFX_LIFESPAN_EXTEND`, `EFX_SOUL_NOURISH`, and `EFX_QI_PURITY_REFINEMENT`.
- [ ] **Fix `_dantianBonusPct` & Dual Lifespan Fields**: Persist `npc.ext.dantianBonusPct` (wired into `CULT.qiMax`) and unify `npc.ext.pillLifespanYears` inside `CULT.refreshLifespan` with diminishing returns + Elder near-end-of-life purchase priority in `CPE.pickPurchase`.
- [ ] **Fix `starter: true` on Tier-7 Partial/Heritage Recipes (`data_items.js`, `economy.js:264, 309`)**: Gate `REC140_PARTIAL_DRAGON_PILL_7` & `REC140_TR_ASTRAL_NIRVANA_ELIXIR` behind discovery/deciphering and cap `+ (r.starter ? 1 : 0)` at `r.tier <= 2`.
- [ ] **Wire `dailyProgressCap`, `Math.max` on `pillBoost`, `insightMult` on `ITEM190_PILL_006`, & Spirit-Fish Alchemy Reserve**.